QP Code **Reg.No : .....**

Name : .....

MODEL QUESTION PAPER

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

Semester IV

MG4DSCMAT202- Applied Mathematical Methods

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. Define algebraic equations. Give an example. [U] / [CO1]
2. Define transcendental equations. Give an example. [U] / [CO1]
3. Find the fundamental period of the function $sin2x$. [U] / [CO2]
4. Draw the graph of the function $f(x) = cos2x$ for $0 \leq x \leq 2\pi$. [U] / [CO2]
5. Write the Laplace transform of $\sin \omega t$. [U] / [CO3]
6. Write $\mathcal{L}(f''(\mathrm{t}))$ in terms of $\mathcal{L}(\mathrm{f(t)}), \mathrm{f(0)}$ and $f'(0)$. [A] / [CO3]
7. Write the general solution of $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$. [U] / [CO4]
8. Write the general form of Lagrange's first order partial differential equation. [K] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **5** questions

Each question carries **6** marks

9. Use iteration method to find a positive root of the equation $x\,e^x = 1$, given that the root lies between 0 and 1. [A] / [CO1]

10. Use Newton Raphson Method to find a root of the equation $x^3 - 2x - 5 = 0$. [A] / [CO1]

11. Write the general Fourier series formula for a periodic function of period 2π. [U] / [CO2]

12. Check whether the function $f(x) = x^3, -\pi < x < \pi$ is even or odd. Write the general Fourier series formula for an odd function. [A] / [CO2]

13. If $\mathcal{L}\big(\mathrm{f(t)}\big) = F(s)$, find $\mathcal{L}\big(e^{at}\mathrm{f(t)}\big)$ and hence find $\mathcal{L}(e^{5t}sint)$. [A] / [CO3]

14. If $F(s)$ is the Laplace transform of $f(t)$, find $\mathcal{L}\left(\int_0^t f(\tau)d\tau\right)$. [A] / [CO3]

15. Solve $\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$. [A] / [CO4]

16. Eliminate the constants a and b from $z = (x + a)(y + b)$ to form the first order partial differential equation. [A] / [CO4]

[6x5 = 30]

Part C

Essay Type Questions

Answer any **3** questions

Each question carries **10** marks

17. Obtain a root, correct to three decimal places of each of the following equations [A] / [CO1]
    (a) $x^3 - x - 4 = 0$
    (b) $1 + x^2 - x^3 = 0$

18. Find the Fourier series for the function [A] / [CO2]

    (a) $f(x) = \begin{cases} -1, & -2 < x < 0 \\ 1, & 0 < x < 2 \end{cases}$, $p = 4$

    (b) $f(x) = \begin{cases} 1 + x, & -1 < x < 0 \\ 1 - x, & 0 < x < 1 \end{cases}$, $p = 2$

19. Check whether the following functions are even or odd. Hence find the corresponding Fourier series for [A] / [CO2]

(a) $f(x) = \pi - |x|, -\pi < x < \pi.$

(b) $f(x) = 1 - \frac{|x|}{2} \quad -2 < x < 2.$

20. (a) Find the Laplace transform of (i) $t\,cos5t$ (ii) te^{kt} [A] / [CO3]

(b) Solve $y'' + 2y' + 2y = 0,\ y(0) = 1, y'(0) = -3$ using Laplace transforms.

21. (a) Find $\mathcal{L}\big(\mathrm{f(t)}\big)$ where $f(t) = t^2 sin3t.$ [A] / [CO3]

(b) Find the inverse Laplace transform of $\ln\left(1 + \frac{\omega^2}{s^2}\right).$

22. Find the general solution of the differential equation [A] / [CO4]

(a) $z(xp - yq) = y^2 - x^2.$

(b) $x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = (x + y)z.$

[15x2 = 30]